

Apply cot both side, we get

$$\cot \theta = \cot \left(\frac{\pi}{2} - \frac{1}{2} \tan^{-1} \frac{4}{3} \right)$$

$$\cot \theta = \tan \left(\frac{\tan^{-1} \frac{4}{3}}{2} \right) \quad \dots(i)$$

Let $\tan^{-1} \frac{4}{3} = \alpha$

We know that $\tan \frac{\alpha}{2} = \operatorname{cosec} \alpha - \cot \alpha$

From (i) we get

$$\cot \theta = \operatorname{cosec} \left(\tan^{-1} \frac{4}{3} \right) - \cot \left(\tan^{-1} \frac{4}{3} \right) \quad \dots(ii)$$

We have $\tan^{-1} \frac{4}{3} = \alpha$

$$\tan \alpha = \frac{4}{3}$$

$$\Rightarrow \cot \alpha = \frac{3}{4} \text{ and } \operatorname{cosec} \alpha = \frac{5}{4}$$

Put in equation (ii) we get

$$\begin{aligned} \cot \theta &= \operatorname{cosec} \alpha - \cot \alpha \\ &= \frac{5}{4} - \frac{3}{4} \\ &= \frac{2}{4} \\ &= \frac{1}{2} \end{aligned}$$

64. Option (c) is correct.

Equation of parabola is $y^2 = 8x$

$$\therefore a = 2$$

We know that focal distance of point $P(x_1, y_1)$ is $x_1 + a$

$$\therefore \text{Focal distance from point } (6, 4\sqrt{3}) \text{ is } 6 + 2 = 8$$

So, statement 1 is correct.

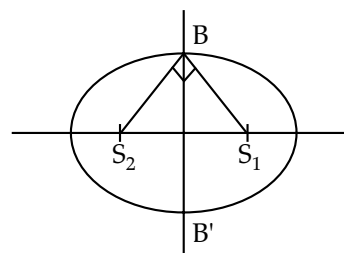
Distance of point $P(6, 4\sqrt{3})$ from directrix

$$\begin{aligned} PF &= \sqrt{(6-2)^2 + (4\sqrt{3}-0)^2} \\ &= \sqrt{16+48} \\ &= 8 \end{aligned}$$

So, statement 2 is also correct.

65. Option (d) is correct.

Let ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$



It's foci are $S_1(ae, 0)$ and $S_2(-ae, 0)$ and the extremity of the minor axis is $B(0, b)$

$$\text{Then slope of } BS_2 = \frac{b-0}{0-(-ae)} = \frac{b}{ae}$$

$$\text{and slope of } BS_1 = \frac{b-0}{0-ae} = \frac{-b}{ae}$$

$\therefore$ The product of slopes, $m_{BS_2} \times m_{BS_1} = -1$

[$\because S_1B$ and S_2B are perpendicular]

$$\Rightarrow \frac{-b}{ae} \times \frac{b}{ae} = -1$$

$$b^2 = a^2 e^2$$

$$\Rightarrow e^2 = \frac{b^2}{a^2} \quad \dots(i)$$

As we know that $e = \frac{c}{a}$, where

$$c = \sqrt{a^2 - b^2}$$

$$\Rightarrow e^2 = 1 - \frac{b^2}{a^2}$$

$$= 1 - e^2 \quad [\text{from (i)}]$$

$$\Rightarrow 2e^2 = 1$$

$$\Rightarrow e^2 = \frac{1}{2}$$

$$\Rightarrow e = \frac{1}{\sqrt{2}}$$

66. Option (d) is correct.

We know that $\vec{a} = \hat{i} - \hat{j} + \hat{k}$ and

$$\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$$

We need to find $\vec{a} \times (\vec{b} \times \vec{a})$

We know that

$$\begin{aligned} \vec{a} \times (\vec{b} \times \vec{a}) &= (\vec{a} \cdot \vec{a})\vec{b} - (\vec{a} \cdot \vec{b})\vec{a} \\ &= [(\hat{i} - \hat{j} + \hat{k}) \cdot (\hat{i} - \hat{j} + \hat{k})]\vec{b} - [(\hat{i} - \hat{j} + \hat{k}) \cdot (\hat{i} + 2\hat{j} - \hat{k})]\vec{a} \\ &= (1+1+1)\vec{b} - (1-2-1)\vec{a} \\ &= 3(\hat{i} + 2\hat{j} - \hat{k}) + 2(\hat{i} - \hat{j} + \hat{k}) \\ &= 5\hat{i} + 4\hat{j} - \hat{k} \quad \dots(i) \end{aligned}$$